

Controller Area Network (CAN) Protocol

Complete Technical Guide for Automotive Embedded Systems

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Target Audience:

- Automotive Engineers
- Embedded Systems Developers
- Students and Researchers
- Technical Professionals

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1. Introduction to CAN Protocol

1.1 What is CAN?

Controller Area Network (CAN) is a robust, multi-master serial bus protocol designed for reliable communication in harsh environments like automotive applications. **Key Characteristics:** • Multi-master capability - any node can initiate communication • Message-based protocol - messages have IDs, not addresses • Priority-based arbitration - critical messages get through first • Excellent error detection - five different error checking mechanisms • Differential signaling - immune to electrical noise

1.2 Historical Development

1983: Robert Bosch begins development 1986: First CAN specification released by Bosch 1991: Mercedes-Benz W140 - first production car with CAN 1993: ISO 11898 standard published 2000s: CAN becomes standard in nearly all vehicles worldwide 2012: CAN FD (Flexible Data-rate) specification released 2015: CAN FD ISO standardized (ISO 11898-1:2015)

1.3 Applications

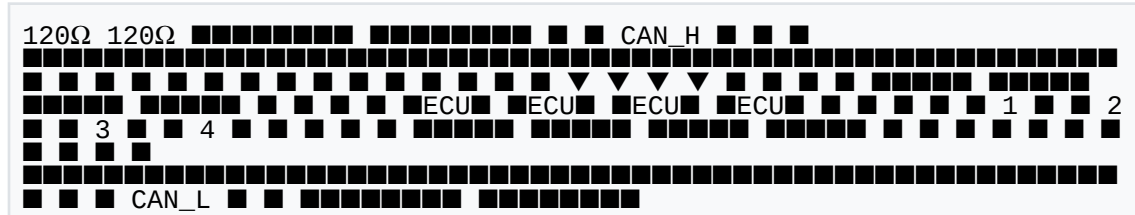
Automotive (Primary Use): • Powertrain: Engine, transmission control • Chassis: ABS, ESP, traction control • Safety: Airbags, seatbelt tensioners • Body: Lights, windows, climate control • Infotainment: Audio, navigation, displays **Other Industries:** • Industrial Automation: PLCs, sensors, robotics • Medical Equipment: Patient monitoring systems • Aerospace: Avionics, UAV control systems • Marine: Navigation and control systems

2. CAN Network Architecture

2.1 Network Topology

CAN uses a linear bus topology with all nodes connected in parallel. Each node has a transceiver connected to the two-wire bus (CAN_H and CAN_L). Critical components:

- **CAN_H (High line)**: One wire of differential pair
- **CAN_L (Low line)**: Second wire of differential pair
- **Termination resistors**: 120Ω at BOTH ends
- **ECUs**: Electronic Control Units with CAN interfaces



2.2 ECU Internal Architecture

Each ECU consists of three main components:

1. **Microcontroller (MCU)** - Application software - Data processing - Control algorithms
2. **CAN Controller** - Protocol implementation - Message buffers (TX/RX) - Error detection - Bit timing control
3. **CAN Transceiver** - Voltage level conversion - Differential signaling - Bus protection - Physical layer interface

